Block 2
Money

Unit 4	Role of Money in Modern Economy
Unit 5	Demand for Money
Unit 6	Money Supply

UNIT 4 THE ROLE OF MONEY IN A MODERN ECONOMY

Structure

4.0 OBJECTIVES

After going through the unit, you will be able to:

- Define money;
- Describe the nature and functions of money;
- Explain measures of money supply;
- Discuss the payment system in a modern economy; and
- Analyse the role of money and credit in the macroeconomy.

4.1 INTRODUCTION

In block-1, you have studied the nature of financial institutions and markets, economic agents in the financial system and also about financial instruments. In this block, we discuss the basic medium of exchange in the economy, namely, money. We will see that money may not be a proper 'financial instrument' but is required for transactions. The present unit, which is the first in the block-2, discusses the basic nature of money. The unit tries to explain what money is, and what is meant by 'near-monies', financial instruments which seem to be quite like money, but do not perform all the functions of money. The unit discusses the functions that money performs and its role in the modern economy. What is the basic structure of payment settlements in a modern economy? How are payments made in transactions across the entire economy? The unit also discusses the basic payment system of an economy that is monetarily and financially mature. Finally, you will be presented with a discussion regarding the role and impact of money, debt and credit in the macroeconomy. The structure of the unit is as follows. Section 4.2 discusses the nature and functions of money. We discuss various measures of Money Supply in Section 4.3. and Section 4.4 describes the payment system of an economy, with the payment system of India used as an illustration. Finally, Section 4.5 discusses the relation between money and credit, and the role they have in the financial system and the macroeconomy.

4.2 NATURE AND FUNCTIONS OF MONEY

Money is anything that is generally acceptable as a means of payment in the settlement of transactions, including debt.

Money can be referred to as one of the greatest innovations of human society. Money is not needed for its own sake as one needs food, clothes and a living house. Money is needed to mediate transactions. It has purchasing power which enables us to exchange goods and services. So, this makes money a unique commodity.

By reducing transaction costs and thereby encouraging more trade, it can be argued that money allows the development of specialization of labour that is necessary for rapid economic growth. Thus, without money, all must be largely self-sufficient as happens in primitive and tribal societies or in barter economy.

Today, we are all familiar with money in our day-to-day transactions. Perhaps the oldest and simplest role of money has been the 'medium of exchange' for all economic transactions as it is acceptable to everybody. Besides this money is the commonly used means of transferring purchasing power. It does not need to be converted into something else before it can be spent or used for the settlement of debt. What makes money 'money' is the belief held by everyone that it will be accepted as such by all others in the economy.

In all money serves **four** important functions for any society: **Medium of Exchange, Unit of Value, Standard of Deferred Payment, and Store of Value.**

Defining Money: Anything that functions as a means of payment (medium of exchange), unit of account, and store of value. Today money also acts as a legal tender which enhances the trust of people in it.

Measuring Money is not as easy as defining it. What, in your opinion, should we use as money?

- Just currency?
- Currency and other stored forms of money?

We find there are multiple definitions of money and what we call money has changed over time as technology and financial instruments have evolved.

Near Money Definition

Highly liquid assets which are not cash but can easily be converted into cash are referred to as near money assets.

Various sources provide the following examples of near money:

- Money market funds
- Bank and Post Office deposits
- Government treasury securities
- Bonds near their redemption date
- Foreign currencies, especially widely traded ones such as the Euro, Dollar, Pound Sterling or Yen
- Equities/shares of the corporate sector

Money has four primary functions in any economy: as a medium of exchange, as a unit of account, as a store of value and standard of deferred value. Of the four functions, its function as a medium of exchange is what distinguishes money from other assets such as stocks, bonds, and houses.

Medium of Exchange Function

Money as a medium of exchange may be used for any transactions wherein goods or services are purchased or sold. Money as a unit of account can be used to value goods or services and express it in monetary terms. Money can also be stored or conserved for future purposes.

In almost all market transactions in our economy, money in the form of currency or cheques is a medium of exchange; it is used to pay for goods and services. The use of money as a medium of exchange promotes economic efficiency by minimizing the time spent in exchanging goods and services.

Let's consider a *barter economy*, one that is without money, in which goods and services are exchanged directly for other goods and services. The time spent trying to exchange goods or services is called a *transaction cost*. In a barter economy, transaction costs are high because people have to satisfy a "double coincidence of wants"— they have to find someone who has a good or service they want and who also wants the good or service they have to offer.

Money promotes economic efficiency by eliminating much of the time spent exchanging goods and services. It also promotes efficiency by allowing people to specialize in what they do best. Money is, therefore, essential in an economy: it allows the economy to run more smoothly by lowering transaction costs, thereby encouraging specialization and division of labour.

The need for money is so strong that almost every society creates it. For a commodity to function effectively as money, it has to meet several criteria: (1) It must be easily standardized, making it simple to ascertain its value; (2) it must be widely accepted; (3) it must be divisible so that it is easy to "make change"; (4) it must be easy to carry; and (5) it must not deteriorate quickly. Objects that have satisfied these criteria have taken many unusual forms throughout human history.

Unit of Account Function

The second role of money is to provide a unit of account; that is, money is used to measure value in an economy. We measure the value of goods and services in terms of money, just as we measure weight in terms of pounds or distance in terms of miles. To see why this function is important, let's look again at a barter economy, in which money does not perform this function. If the economy has only three goods X, Y, Z then we need to know only three prices to tell us how to exchange one for another: the price of X in terms of Y (that is, how many Ys one has to pay for an X), the price of Y in terms of Z and the price of Z in terms of X. As the number of goods increases in this economy the number of relative prices goes up tremendously.

The solution to the problem is to introduce money into the economy and have all prices quoted in terms of units of that money. We can see that using money as a unit of account lowers transaction costs in an economy by reducing the number of prices that need to be considered. The benefits of this function of money grow as the economy becomes more complex.

Store of Value Function

The store-of-value function of money pertains to its capacity to retain purchasing power over time. In other words, money serves as a means to hold wealth in a stable form that can be readily accessed and utilized in the future. People use money to preserve the economic value of their assets and savings.

However, the effectiveness of money as a store of value can fluctuate due to factors such as inflation, economic stability, and the availability of alternative investment options. Inflation, for example, can diminish the value of money over time, reducing its purchasing power. In such scenarios, individuals might seek alternative stores of value, like real estate, precious metals, stocks, or bonds, which may offer better protection against inflation or higher returns.

Despite these challenges, money remains a widely accepted and convenient store of value because of its liquidity, divisibility, and broad acceptance of goods and services. Although it may not always preserve wealth as effectively as other assets, its ease of use and stability make it a crucial element of modern economies.

Check Your Progress 1

1) Discuss some limitations of the Barter system.

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2) What is Near money? Give some examples of Near money.

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3) Briefly explain the various functions of Money.

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4.3 MEASURES OF MONEY SUPPLY

There is a vast array of financial assets in any economy, from currency to complicated claims on other financial assets. Which of these assets should be called money? There are two types of measures of money: narrow money and broad money. Narrow money comprises those claims that can be used directly, instantly, and without restrictions to make payments. These claims are *liquid*. Broad money, includes, in addition, claims that are not instantly liquid — withdrawal of time deposits, for example, may require notice to the depository institution; money market mutual funds may set a minimum on the size of cheques drawn on an account.

The money supply is the total amount of money available in an economy at a particular point in time (a stock).

Usually, the money stock in an economy is measured as one of the following alternative monetary aggregates:

1) M0; i.e., the monetary base, alternatively called base money, central bank money, or high-powered money. The monetary base is defined as fully liquid claims on the central bank held by the private sector, that is, currency (coins and notes) in circulation plus demand deposits held by the commercial banks in the central bank.

   This monetary aggregate is under the direct control of the central bank and is changed by open-market operations, that is, by the central bank trading bonds, usually short-term government bonds, with the private sector. However, the monetary base is an imperfect measure of the liquidity in the private sector. We will discuss more about high-powered money in Unit 6.

2) M1: it is defined as currency in circulation plus demand deposits held by the non-bank general public in commercial banks. M1 does not include currency held by commercial banks and demand deposits held by commercial banks in the central bank. Yet M1 includes the major part of M0 and is generally considerably larger than M0.

Broader categories of money include:

3) M2: M1 plus savings deposits with unrestricted access i.e., demand deposits. Although these claims may not be instantly liquid, they are close.

4) M3: M1 plus time deposits. M3 captures even broader liquidity in the economy.

5) M4: This measure includes M3 plus other forms of broad money, which may include some longer-term deposits and other less liquid financial assets like post office deposits.

4.4 MONEY AND THE PAYMENTS SYSTEM

The journey of money through history is intriguing, stretching back thousands of years. It began with simple barter systems, where goods and services were exchanged directly without a standardized medium. However, the limitations of barter, such as the difficulty in finding matching needs, led to the creation of more efficient methods of exchange.

The first significant innovation came with the introduction of metal coins. Early coins were made from bronze and copper, and later, more valuable metals like silver and gold were used. These coins represented a breakthrough because they allowed people to make payments by count rather than by weight, which simplified transactions and enabled broader commerce.

As societies and economies grew more complex, the use of coins evolved further. Alongside the physical exchange of coins, people began to develop financial instruments like claims on goods and bills of exchange. These instruments facilitated trade by providing a promise of payment or transfer of goods at a later date, thereby adding flexibility and trust to economic transactions.

The next monumental step in the history of money was the creation of paper money. Paper currency offered the advantage of being lighter and easier to transport than metal coins, which was particularly useful for large transactions and long-distance trade. This innovation marked a significant shift in how wealth was stored and transferred, and it laid the groundwork for modern banking systems.

In the present day, payment systems are undergoing rapid transformation due to technological advancements. Users now expect payments to be faster, easier, and accessible at any time, mirroring the digitalization and convenience of other aspects of life. This demand has led to the widespread adoption of near-instant person-to-person retail payment systems. These systems leverage digital technology to enable quick and seamless transactions, reflecting the continuing evolution of money and its pivotal role in society. The most recent and revolutionary phase in the evolution of money is the advent of cryptocurrency. Cryptocurrencies, such as Bitcoin and Ethereum, are digital or virtual currencies that use cryptography for security and operate on decentralized networks based on blockchain technology. They challenge traditional banking and financial systems by providing alternative ways of storing and transferring value. However, cryptocurrencies are often criticized for their extreme price volatility, regulatory uncertainty, security risks, scalability issues, energy consumption, limited acceptance, technical complexity, legal and tax implications, market manipulation, and lack of consumer protection.

Overall, the evolution of money from barter to digital payments highlights how financial innovations have consistently shaped and facilitated economic and societal development. As we move forward, the pace of change in payment systems is likely to accelerate, driven by ongoing technological advancements and the increasing expectations of consumers for convenience and speed.

Payment Mechanism-A Case of India

India also has consistently promoted innovation and development in payment and settlement systems. With the developments in the economy and the evolution of the payment system, the form and functions of money have changed over time, and it will continue to influence the future course of currency. The concept of money has experienced an evolution from commodity to metallic currency to paper currency to digital currency. The changing features of money are defining the new financial landscape of the economy. Over the past decades, numerous payment systems have emerged to benefit the common man. Since the mid-eighties, the Reserve Bank of India has introduced various technology-based solutions to enhance the banking system.

In a country like India, modern payment systems can help foster economic development and financial stability as well as support financial inclusion. The past decade has also witnessed the blossoming of quite a few payment systems that are affordable, accessible, convenient, efficient, safe, secure and available 24×7×365 days a year.

Alongside innovations in digital payments, India has implemented a separate law for Payment and Settlement Systems, to ensure the structured development of the country's payment ecosystem. The significant shift in payment preferences has resulted from the establishment of robust, roundthe-clock electronic payment systems like Real Time Gross Settlement (RTGS) and National Electronic Funds Transfer (NEFT), which facilitate seamless real-time or near real-time fund transfers. Additionally, the introduction of Immediate Payment Service (IMPS) and Unified Payments Interface (UPI) for instant payment settlements, along with mobile-based payment systems such as Bharat Bill Payment System (BBPS) and National Electronic Toll Collection (NETC) for electronic toll payments, have been transformative milestones. These advancements have attracted international recognition and ensured rapid acceptance due to the convenience they offer, providing consumers with alternatives to cash and paper-based payments. The involvement of non-bank FinTech firms as PPI issuers, Bharat Bill Payment Operating Units (BBPOUs), and third-party application providers on the UPI platform has further promoted the adoption of digital payments in India.

Furthermore, with the advent of cutting-edge technologies, the digitalization of money is the next milestone in monetary history. Advancement in technology has made it possible for the development of a new form of money viz. Central Bank Digital Currencies (CBDCs). Given the volatility and other limitations of cryptocurrency, recent innovations in technology-based payment solutions have led central banks around the globe to explore the potential benefits and risks of issuing a CBDC to maintain the continuum with the current trend in innovations. RBI also is currently engaged in working towards a phased implementation strategy, going step by step through various stages of pilots followed by the final launch, and simultaneously examining use cases for the issuance of its own CBDC (Digital Rupee (e₹)), with minimal or no disruption to the financial system. Currently, India is at the forefront of an inflexion point in the evolution of currency that will decisively change the very nature of money and its functions.

Reserve Bank broadly defines CBDC as the legal tender issued by a central bank in a digital form. It is akin to sovereign paper currency but takes a different form, exchangeable at par with the existing currency and shall be accepted as a medium of payment, legal tender and a safe store of value. CBDCs would appear as a liability on a central bank’s balance sheet. The Bank for International Settlement has laid down “foundational principles” and “core features” of a CBDC, to guide exploration and support public policy objectives, as per the need of the existing mandate of Central Banks. The foundational principles emphasise that authorities would first need to be confident that issuance would not compromise monetary or financial stability and that a CBDC could coexist with and complement existing forms of money, promoting innovation and efficiency. Supported by state-of-the-art payment systems of India that are affordable, accessible, convenient, efficient, safe and secure, the Digital Rupee (e₹) system will further bolster India’s digital economy, make the monetary and payment systems more efficient and contribute to furthering financial inclusion.

The dynamic and accelerated development of the payments ecosystem in India, facilitated by increased adoption of technology and innovation, has established the country as a force to reckon with in the global payments space, in terms of not only growth in digital payments but also the availability of a bouquet of safe, secure, innovative and efficient payment systems. Over 26 crore digital payment transactions are processed daily by our payment systems, of which the Unified Payments Interface (UPI) system itself processes more than two-thirds.

Check Your Progress 2

1) How has money evolved over the years? Highlight the role of electronic money and mobile payment systems in modern transactions.

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2) Discuss the evolution of payment mechanisms in India and recent innovations in the field.

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3) What is Central Bank digital currency? Discuss some of its potential benefits.

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4.5 MONEY, CREDIT AND THE MACROECONOMY

Understanding the intricate dynamics between money, credit, and the macroeconomy is essential for identifying how economic policies influence aggregate demand, output, and price levels.

When the demand for money rises, it reduces aggregate demand unless the monetary authority recognizes the increase in time and push up the money supply by an equal amount. Understanding money demand, and how various factors affect that demand, is, therefore, a first step in setting a target for the monetary authority. Money demand refers to the desire to hold cash or liquid assets instead of investing in less liquid assets. This demand is influenced by several factors, including income levels, interest rates, and economic uncertainty. Higher-income increases the transaction demand for money, while higher interest rates typically reduce the demand for money since holding money becomes more costly compared to earning potential returns from other investments.

When the demand for money rises, it leads to higher interest rates. Higher interest rates can dampen investment and consumer spending, reducing aggregate demand. This reduction in aggregate demand can slow down economic growth and potentially lead to a recession if not addressed promptly. We will explore the concept of demand for money in greater details in Unit 5: Demand for Money.

To counteract the negative impact on aggregate demand, the central bank can increase the money supply. Aggregate demand rises when money supply increases faster than money demand — with a concomitant rise in output or the price level.

By injecting more money into the economy, the central bank can stabilise interest rates and support aggregate demand. This can happen through mechanisms like open market operations, where the central bank buys government securities, thereby increasing the reserves of commercial banks and enabling them to lend more. We will discuss more about various instruments of monetary policy in Block 3.

An increase in the money supply typically lowers interest rates, making borrowing cheaper for businesses and consumers. Lower interest rates reduce the cost of financing investments and major purchases, leading to higher spending on capital goods and consumer products. As businesses expand and consumers increase their expenditure, aggregate demand rises, boosting economic output and employment. However, if the economy is already near full capacity, the increased demand may primarily lead to higher prices, causing inflation.

This intervention requires timely recognition of changes in money demand to avoid economic instability. If the monetary authority fails to recognize and respond to this shift promptly, the economy may experience slower growth or even a contraction.

The Transmission Mechanism of Monetary Policy: The Credit Channel

Ben Bernanke's seminal work "Credit in the Macroeconomy" explores the critical role of credit in influencing economic activity and macroeconomic stability. Bernanke's analysis highlights how fluctuations in credit availability can significantly impact economic performance, often through channels that traditional monetary theories might not fully account for.

Bernanke emphasizes the importance of the "credit channel" of monetary policy, which operates alongside the traditional interest rate channel. The credit channel affects the economy through the availability and terms of credit, impacting businesses' and consumers' ability to borrow and spend. This is especially relevant during periods of financial distress when credit markets tighten, leading to a reduction in investment and consumption, and consequently, a slowdown in economic activity.

Bernanke and Blinder (1988) highlighted how monetary policy impacts bank lending behaviour, establishing a foundational understanding of the bank lending channel. Bernanke and Gertler (1995) explored the balance sheet channel, showing how changes in firms' net worth influence their investment decisions and the broader economic impact. Recent studies have shown that during the 2008 financial crisis, the impaired credit channel significantly contributed to the depth and duration of the recession (Ivashina and Scharfstein, 2010).

The credit channel is particularly relevant in the presence of asymmetric information in financial markets. Banks and lenders have more information about the creditworthiness of borrowers than external investors, making bank lending a crucial source of finance for many firms and households. (Bernanke, 1993)

However, it is also important to note that the effectiveness of the credit channel can vary over the business cycle. During economic downturns, even with expansionary policy, banks may be hesitant to lend due to increased risk perceptions, limiting the channel's effectiveness. Conversely, during booms, easier credit can exacerbate economic expansions.

The credit channel can be broken down into two main components:

The Bank Lending Channel

The Balance Sheet Channel

Both components emphasize the role of financial intermediaries (banks) and the financial health of borrowers in the transmission of monetary policy.

1) **The Bank Lending Channel**

The bank lending channel operates through the impact of monetary policy on banks' ability to lend. We have already discussed it earlier how when the Central Bank implements expansionary monetary policy (e.g., lowering the interest rate or purchasing securities), it increases the reserves of commercial banks. Then, the banks have more funds available to lend. As the cost of funds decreases, banks can offer more loans at lower interest rates. With more accessible and cheaper credit, businesses and consumers are more likely to borrow for investment and consumption. This increased borrowing can lead to higher aggregate demand, stimulating economic growth. Conversely, contractionary monetary policy (e.g., raising interest rates) reduces bank reserves, leading to tighter lending conditions, higher borrowing costs, and lower aggregate demand.

2) **The Balance Sheet Channel**

The balance sheet channel focuses on how monetary policy affects the financial position of borrowers. Expansionary monetary policy often leads to higher asset prices (e.g., stocks, real estate). As asset values increase, the net worth of businesses and households improves. Higher net worth enhances the value of collateral that borrowers can pledge, improving their borrowing capacity. With better financial health, borrowers face lower borrowing costs and more favourable loan terms. Improved balance sheets make businesses more willing to invest in new projects and consumers more inclined to spend, further boosting economic activity. In a contractionary scenario, falling asset prices reduce net worth and collateral values, restricting borrowing capacity, and dampening investment and consumption.

Check Your Progress 3

1) Briefly describe the relationship between money, credit and macroeconomic policy.

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2) Explain the credit transmission channel of monetary policy.

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4.6 LET US SUM UP

The unit provided a discussion about the definition, nature and functions of what is commonly called 'money'. We saw that anything that is universally accepted as a medium of exchange is called money. The unit briefly discussed the disadvantages of a barter economy, and how the introduction of money facilitates exchange and reduces transaction costs. The unit also explained the meaning of the concept of 'liquidity'. We saw money is the most 'liquid' of instruments, and some other instruments, which perform some of the same functions that money does, are not as 'liquid' as money and hence are called 'near monies.

The unit discussed the functions of money. We saw that money serves as a medium of exchange, a unit of account, a store of value, and a standard of deferred payments. This last function which allows a transaction and its payment to be separated in time, is the source of credit and debt in the economy. The unit brought out the difference between money, income, wealth and assets. The system of payments in a modern economy was discussed and the means and methods of payments were listed. Elements of the macroeconomic relationship between money and credit and various macroeconomic variables were discussed.

4.7 KEY WORDS

Barter	:	A system where goods and services are exchanged for other goods and services.
Near Money	:	Highly liquid assets which are not cash but can easily be converted into cash.
Cryptocurrency	:	Digital or virtual currencies that use cryptography for security and operate on decentralized networks based on blockchain technology.
Monetary Policy	:	Monetary policy is the policy adopted by the monetary authority of a nation to affect monetary and other financial conditions to accomplish broader objectives like high employment and price stability.
Digital Currency	:	Digital currency is any currency, money, or money-like asset that is primarily managed, stored or exchanged on digital computer systems.
Unified Payments Interface (UPI)	:	Unified Payments Interface (UPI) is a real-time payment system in India that enables seamless money transfers from one bank account to another instantly and free of charge through a mobile device.
Real Time Gross Settlement (RTGS)	:	Real-time gross settlement systems are specialist funds transfer systems where the transfer of money or securities takes place from one bank to any other bank on a "real-time" and on a "gross" basis to avoid settlement risk.
National Electronic Funds Transfer (NEFT)	:	The NEFT or National Electronic Funds Transfer is an electronic payment system that allows users to initiate direct one-to-one payment anywhere across the country. One can send money to the beneficiary only if he or she has a bank account with any branch in the country.

4.8 SOME USEFUL REFERENCES

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4.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

1) Refer to Section 4.2

2) Refer to Section 4.2

3) Refer to Section 4.2

Check Your Progress 2

1) Refer to Section 4.4

2) Refer to Section 4.4

3) Refer to Section 4.4

Check Your Progress 3

1) Refer to Section 4.5

2) Refer to Section 4.5

UNIT 5 DEMAND FOR MONEY

Structure

5.0 OBJECTIVES

The objectives of the units are as follows:

- Learning the meaning of demand for money;
- Understanding the main factors affecting the demand for money; and
- Understanding various economic theories of demand for money.

5.1 INTRODUCTION

Money demand plays a crucial role in understanding the broader functioning of economies, influencing monetary policy decisions, and shaping financial markets. We start our exploration with the meaning of money demand curve and the multifaceted factors that influence money demand, from interest rate to technological advancements. These factors lay the groundwork for our deep dive into various theories of money demand, each offering unique insights into this fundamental aspect of economic behaviour. First, we will explore the classical Quantity Theory of Money followed by the Cambridge version. Next, we will delve into Keynes' Liquidity Preference Theory, which emphasizes various motives of holding money and the role of interest rates in shaping individuals' decisions to hold money versus other assets. We will also examine the Baumol-Tobin theory for transaction demand, which focuses on the transaction motive for holding money balances and introduces the concept of opportunity cost in deciding between holding money and holding interest-bearing assets. Furthermore, we will explore Tobin's theory of speculative demand for money, which sheds light on how the portfolio of cash and bonds is optimized and its relationship with the speculative demand for money. Finally, the unit explores the Friedman's quantity theory of money. By the end of this unit, you will gain a comprehensive understanding the demand for money equipping you with valuable insights into the functioning of an economy.

5.2 MONEY DEMAND

Money demand means the quantity of money (or cash) which households and firms want to hold with themselves, but not in other forms like bonds. Keeping other factors constat, just as the demand for a good is negatively related with its price, similarly, the demand for money is negatively related with interest rate. Therefore, a money demand curve (showing the relationship between interest rate and the money demand) slops downward as shown in Figure 5.1. The simple reasoning behind this negative slope is that holding money at home does not earn anything and it loses the interest being offered by other assets like government bonds if that money is invested in the bonds. Thus, interest rate is the opportunity cost of holding or demanding money.

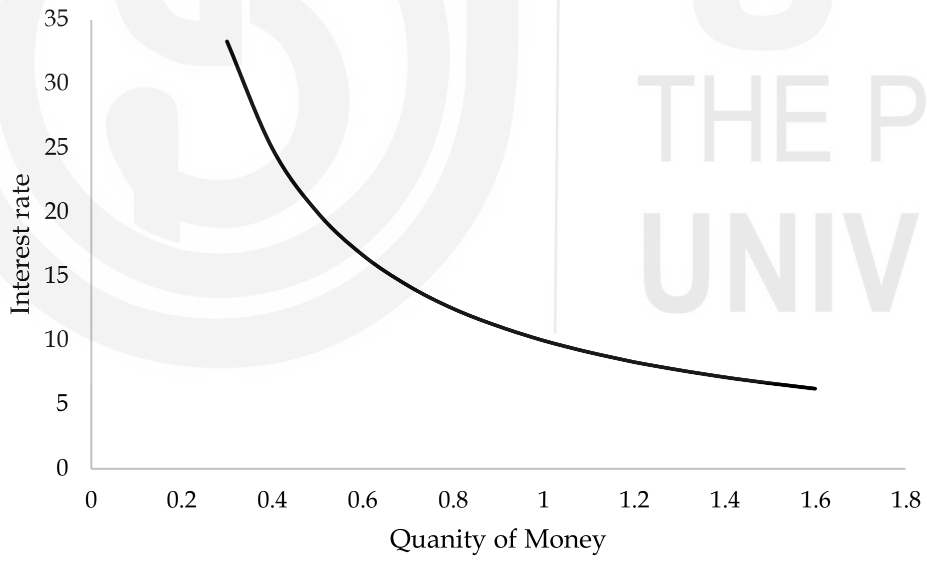

Fig. 5.1: Money Demand Curve

Check Your Progress 1

1) What is money demand?

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2) Explain the money demand curve.

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5.3 FACTORS AFFECTING THE DEMAND FOR MONEY

The main factors affecting the demand for money are as follows:

1) Interest rate

Keeping other factors constant, the demand for money is influenced by the anticipated returns of both money and alternative assets. When the expected return on money rises, the demand for money increases. Conversely, when the expected return on alternative assets increases, individuals with wealth tend to shift from holding money to investing in these higher-return alternatives, resulting in a decrease in the demand for money. For example, suppose that an investor who has a certain amount of wealth to allocate between holding money and investing in stocks. Suppose initially, the expected return on both holding money (such as keeping it in a savings account) and investing in stocks is 5%.

Now, let us suppose that there is a shift in the market, and the expected return on stocks increases to 8%. In this scenario, the investor may decide to reallocate some of their wealth from holding money to investing in stocks. This decision is driven by the desire to earn a higher return on their investment.

Conversely, if the expected return on holding money increases to 7%, while the expected return on stocks remains at 5%, the investor might decide to hold more money rather than investing in stocks. This is because the higher return on holding money makes it a more attractive option compared to investing in stocks.

In both cases, the investor's decision to allocate their wealth between holding money and investing in alternative assets is influenced by the expected returns of both options, illustrating the principles outlined in the theory of portfolio allocation.

2) Price level

When prices rise, people require more currency (e.g., rupee or dollar) to carry out transactions, leading to a greater desire to hold onto money. Let us consider an example: Imagine a small island economy where the price level for basic goods like fruits and vegetables is very low compared to a modern city. In this island economy, let's say a basket of fruits costs only a few island dollars. Now, fast forward a few decades, and due to economic development and inflation, the price level on the island has increased significantly. That same basket of fruits now costs several times more island dollars. As a result of this price increase, the islanders find themselves needing to hold more island dollars to conduct their daily transactions. Their nominal demand for money increases because they require a larger quantity of currency to purchase the same goods and services as before. This increase in the price level directly impacts the amount of money people wish to hold, highlighting the proportional relationship between the price level and the nominal demand for money.

3) Income (or GDP)

Households and businesses retain money to streamline their transactions for goods and services. As their purchasing activity expands, so does their desire to hold more money at any specific interest rate. Consequently, a rise in real GDP, reflecting the overall quantity of goods and services exchanged in the economy, changes the money demand.

4) Risk Associated with Alternative Assets

If the risk associated with alternative assets like stocks and real estate significantly rises, individuals may seek safer options, including holding onto money. Consequently, an increase in overall economic risk may lead to a higher demand for money. Suppose there is a period of economic instability where the stock market experiences a sharp decline, and there is uncertainty regarding the future performance of real estate investments. During such times, investors might perceive stocks and real estate as riskier assets due to the volatility and uncertainty in the market. In response to this increased risk in alternative assets, individuals may choose to shift their investment strategy towards safer options. They might decide to hold onto more cash or invest in low-risk assets like government bonds or savings accounts, which offer stability and security during uncertain times. This increased preference for safer assets, including holding onto money, illustrates how changes in the riskiness of alternative investments can influence the demand for money.

5) Liquidity of Alternative Assets

As alternative assets become more readily convertible into cash, the necessity of holding money diminishes. Similarly, if the liquidity of alternative assets is less, then the demand for money increases.

6) Payments and Receipts Technology

UPI (Unified Payments Interface) platforms such as PhonPe, Google Pay, Paytm have substantially reduced the demand for physical cash in India. Similarly, net banking system of payments and receipts have also reduced the need to carry money in hands. Offering instant, secure transactions directly from bank accounts via smartphones, they provide unparalleled convenience. Moreover, UPI’s accessibility extends even to remote areas with limited banking infrastructure. Enhanced security features instill confidence in users, while incentives like cashbacks further incentivize digital transactions. Integration into various businesses and services ensures widespread acceptance, encouraging consumers to opt for digital payments over cash. Consequently, UPI has transformed India's payment landscape, making digital transactions the norm and significantly decreasing reliance on physical currency.

5.4 THEORIES OF MONEY DEMAND

Over a period of time various theories explaining money demand have been developed. These theories are explained here:

5.4.1 Quantity Theory of Money

Although, the quantity theory of money is basically about the money supply not the demand for money, but it can be used to find out the demand for money in equilibrium when the demand for money is equal to its supply. Therefore, we first understand the quantity theory and then will relate it to the demand side.

According to N.G Mankiw, David Hume is credited with the development of the quantity theory of money. However, American economist Irving Fisher is also credited with the algebraically development of the theory. The quantity theory starts with the *transaction version* of the theory which is explained below: As per this version the following equation holds good:

$$M \times V = P \times T$$

Where,

M = Quantity of money i.e., money supply

V = Transaction velocity of money i.e., it shows how many times a dollar or a rupee changes hands in a period of time.

P = Price of a transaction

T = Number of the transactions in a period of time

The above equation is known as the transaction equation or quantity equation. To understand the meaning of the equation we consider the following hypothetical example:

Suppose the quantity of money is ₹ 25 and there are 50 transactions in a year with price of ₹ 5 per transaction, then as per the above equation:

25 V = 5 × 50 = ₹ 250

$$V = \frac{250}{25} = 10$$

The number 10 shows that ₹ 25 must change hands for 10 times to arrange the transaction value of ₹ 250. This is clear that the calculation of T in real life is almost impossible and therefore, the above equation does not seem to be practical. Therefore, we have another version of the quantity theory i.e., *income version*. As per this version the transaction is converted to income equation by replacing T with income Y assuming that Y is a proxy of T. The logic behind this proxy is that T and Y are positively related though they are not equal, yet they are proportional. Thus, the quantity equation is

$$\mathrm{M} \times \mathrm{V} = \mathrm{P} \times \mathrm{Y}$$

Where,

M = Quantity of money i.e. money supply

V = Income velocity of money instead of transaction velocity.

P = Price level

Y = Real output

Relationship Between the Quantity Theory of Money and Demand for Money

From the transaction version of the theory, we know that

$$M \times V = P \times T$$

Think that there is a relationship between the volume of transactions T and the demand for money because the more value of T means more transactions and more transactions means the more demand for money. In this regard, Laidler[1] says that

> " though not stated as such by Fisher, this is equivalent to the following theory of the money market put in supply and demand terms. The demand for money depends upon the volume of the transactions to be conducted in the economy and is equal to a constant fraction of these transactions. Furthermore, the supply of money is given, and in equilibrium the demand for money must be equal to its supply" .

Thus, M can be replaced by M_d in equilibrium situation, therefore, the above equation should be

$$M_d \times V = P \times T$$

Moreover, if V and T are constants represented by $\bar{V}$ and $\bar{T}$, therefore,

$$M_d \times \bar{V} = P \times \bar{T}$$

Or

[1] Laidler, D. E. W. (1977). The Demand for Money: Theories and Evidence (2nd ed.). Dun-Donnelley Publishing Corp.

$$M_d = P \times \frac{1}{\bar{\bar{V}}} \times \bar{T}$$

Let

$$\frac{1}{\bar{\bar{V}}} = k$$

Therefore,

$$M_d = P \times k \times \bar{T}$$

$$M_d = k\ (P \times \bar{T})$$

Thus,

$$\text{Money demand } = k \times \text{volume of transaction}$$

Moreover, in classical frame work, we know that volume of transaction is equal to the Y, therefore,

$$\text{Money demand } = k \times Income$$

Criticism of the quantity theory

a) British economist J.M. Keynes criticised the quantity theory of money in his book *The General Theory of Employment, Interest and Money*. He says that inflation is not a result of money supply but the aggregate demand. When at full employment level, the aggregate demand exceeds the aggregate supply, then there is inflationary gap. The increase in aggregate demand is independent of the money supply change.

b) Nobel laureate Paul Krugman empirically found that the theory is not applicable during in a liquidity trap. Liquidity trap is a situation when the interest reaches to its minimum and further fall in the interest is not expected and monetary policy is ineffective.

c) Austrian-American economist Ludwig von Mises criticised the theory by saying that the quantity theory of money does not focus on money demand side. That's a different matter that in equilibrium, the theory can help to determine the demand for money, but not in non-equilibrium situation.

d) Some economists argue that the quantity theory of money is not theory in actual but the tautology.

5.4.2 The Cambridge Approach or Neoclassical Theory

Neoclassical theory is an improved version of the quantity theory of money. The credit of this theory goes to Alfred Marshall, A.C. Pigou, D.H. Robertson and J.M, Keynes. Since, all these economists were from the Cambridge University, therefore, the theory got popular as the Cambridge version of the quantity theory of money. These Cambridge economists argue that money is mainly demanded for transaction purposes and people do not hold their incomes entirely in money form (i.e., cash at home) because money at home will not earn interest. Thus, only a portion (denoted by k) of the income is held at home for transaction purposes. In equation form,

$$M_d = k(PY)$$

where

Y = Real income

P = Price level

M_d = Money demand

Thus, according to Neoclassical theory of money, demand for money is directly determined, and money supply can be determined in equilibrium situation by keeping M_d equal to money supply. This is an opposite to the quantity theory of money in which money supply is replaced by money demand. Moreover, here k is known as Cambridge k, which is the reciprocal of the k in the quantity theory of money.

5.4.3 Liquidity Preference Theory or Keynes Theory of Demand for Money

The Liquidity Preference Theory, proposed by John Maynard Keynes, is a cornerstone of modern macroeconomics. It suggests that the demand for money is not just determined by transactions but also by the desire of individuals to hold liquid assets for unforeseen contingencies. Keynes accepts that the main motive of money is its use as medium of exchange, but he also recognises that there are two other motives too. According to Keynes, individuals have a preference for liquidity, meaning they value money not only for its ability to facilitate transactions but also for its security and ease of conversion into other assets. According to Keynes, money demand M(d) is the summation of the following three components:

a) Transaction motive M_d^T

b) Precaution motive M_d^p, and

c) Speculative motive M_d^{sp}.

The detail of each component is as follows:

a) Transaction motive

People do various small and big transactions in their day-to-day life. For example: they purchase clothes, they buy cars, they buy cereals, they buy flour, they pay interest, they repay their loans, they pay salaries, they pay electricity bills, etc. All these transactions require money and this demand is met by the income. Keynes confined the transaction motive as the necessity of cash to fill the gap between receipts and planned regular payments (e.g., school fees, electricity bills). Higher the level of income means the higher the level of transactions for goods and services requiring more money. Therefore, the transaction demand is directly related to the income. In equation form:

$$\text{Nominal money demand} = M_d^T = \alpha(PY)$$

Or

$$\text{Real money demand} = \frac{M_d^T}{P} = \alpha Y$$

where, α shows how much of the nominal income (PY) is demanded for transaction purposes. The above equation shows that there is positive relationship between the money demanded for transaction purpose and the nominal income. Keynesians argue that money demand for transaction purpose is interest-inelastic i.e., interest rate does not affect the money demand, however, some economists argue that beyond a certain level of interest rate (i_3 in the diagram), the money demand gets interest-elastic because such high interest rate is perceived to be high opportunity cost of holding cash balance.

b) Precautionary motive

Future is uncertain. Nobody knows what is going to happen in future, therefore, to meet such unforeseen happenings people demand money. Precautionary motive is one of the reasons people save their income to be used in odd circumstances like death of a family member. The demand for precautionary purpose is also directly related to income. In equation form:

$$\text{Nominal money demand} = M_d^p = \beta(PY)$$

Or

$$\text{Real money demand} = \frac{M_d^p}{P} = \beta Y$$

Here, *b* is a constant that shows how much portion of the nominal income (PY) is demand for precautionary purpose. Keynesians think that precautionary purpose demand is also interest-inelastic but some other economists think that beyond a certain level of interest rate it will be interest-elastic. Thus, the behaviour of the precautionary purpose demand curve is the same as that of the transaction purpose demand.

c) Speculative Motive

Although, speculation can be found in any market whether financial market or otherwise, Keynes' explanation for speculation demand for money revolves around bond market. Speculation means making profits due to price fluctuation of an asset (e.g., a share, or bond). For example, if you purchase a bond for ₹ 1000 today and the price reaches to ₹ 1200 after a year, then you will earn ₹ 200 as profit due to price change. Apart, ₹ 200, you will earn some interest also (generally known as coupon payment) for holding the bond for 1 year. The profits due to price fluctuations are known as capital gains. With regard to a perpetual bond (i.e., a bond having infinity maturity period, the value of a bond is calculated using the following formula:

$$V = \frac{C}{i}$$

where,

V = Value of a bond.

C = Coupon i.e., interest on the bond.

i = Market interest rate or discounting rate.

The formula states that there is an inverse-relationship between the value of a bond and the interest rate. As the interest rate rises, the value of the bond falls down and as the interest rate falls, the value rises up. Keynes argues that, at any time, there is a level of interest rate called normal interest rate and as the interest rate goes above it, then an individual expects the interest rate to fall and when the interest rate is below it, then the individual expects the rise in interest rate. When an individual expects the interest rate to fall (known as a bearish expectation), then he will earn capital gain and interest due to holding period of the bond. When he expects the interest rate to rise (known as a bullish expectation), then he will suffer capital loss, but still he will earn coupon (because the coupon is fixed by the issuer). As long as the coupon exceeds the capital loss, then he will hold the bond and all the wealth will be converted into bonds. If the capital loss exceeds the coupon, then he will not hold bonds, but he would prefer his total wealth into money form i.e., cash. At aggregate level, the weightage of each individual's holdings (in money and bond) is not significant and the aggregative speculative demand for money is a smooth function of the interest rate. According to Laidler, the speculative demand for money function can be shown by the following equation where W is real wealth.

$$M_d^{sp} = (P \times W) \times f(i)$$

Therefore, total money demand can be expressed

$$M_d = \alpha Y + \beta Y + (P \times W) \times f(i)$$

Alternatively,

$$M_d = (\alpha + \beta)Y + (P \times W) \times f(i)$$

Thus, money demand is a function of Y, W, and interest rate.

Some textbooks (e.g., Principles of Macronomics by N.G. Mankiw) summaries the liquidity preference theory by the following equation:

$$\text{Demand for nominal money} = M_d = PL(i, Y)$$

Alternatively,

$$\text{Demand for real money} = \frac{M_d}{P} = L(i, Y)$$

Criticism of liquidity preference theory

a) Keynes' segmentation of the demand for money into three categories is impractical as it assumes individuals maintain distinct purses for different purposes, whereas in reality, people typically manage their finances from a single purse that serves all their needs.

b) Some economists have questioned the empirical validity of liquidity preference theory, arguing that it fails to explain observed fluctuations in interest rates and money demand accurately. Empirical studies have shown that factors beyond liquidity preference, such as inflation expectations, economic growth, and central bank policies, also influence interest rates and money demand.

c) The theory assumes that all individuals have the same expectations about future interest rates and income. However, in reality, people have diverse expectations, leading to variations in their demand for money. This heterogeneity is not adequately accounted for in liquidity preference theory.

5.4.4 Baumol – Tobin Theory of Transaction Demand for Money

This theory was developed by William Jack Baumol (1952) and James Tobin (1956) independently, however, their results were found to be similar, that is why, this theory is commonly termed as Baumol – Tobin theory for transaction demand for money. The theory is also known as known as the Baumol-Tobin Model of Cash Management. The theory is similar to the concept of economic order quantity (EOQ) in inventory management. Just as the economic order quantity is the quantity of the material which minimizes the total cost consisting of purchase cost, order cost, and carrying cost, similarly, the Baumol-Tobin theory helps us to find the level of cash (demand for money for transaction purpose) which an individual should hold to minimise total cost consisting of transaction cost and operational cost. How? Although, the theory can be easily understood by minimizing the cost of demand for money with reference to investment in bonds, yet to keep the theory simple, we assume that an individual is willing to spend ₹ Y annually by withdrawing from his bank account carrying interest *i*.

If ₹ Y are withdrawn in one shot at the beginning of the year and spent gradually at constant rate, then at the end of the year the balance should be zero, therefore, his average holding should be Y/2 = [(opening balance + closing balance)/2].

If he makes half withdrawal at beginning of the year, the opening balance should be Y/2 and it will be spent over the first 6 months and another half balance will be withdrawn and spent over the next 6 months, therefore, the average holding should be Y/4. On the same pattern, if he makes Q number of trips to the bank, then at the beginning he withdraws Q/N money, then the average holding should be Q/2N.

Now, to withdraw the cash from bank, he will have to incur two types of costs given below:

a) Cost of transaction – It includes the cost incurred to conclude the transaction of withdrawing cash e.g., travel cost incurred to visit the bank.

b) Opportunity cost – This is nothing but the interest lost, when the cash is withdrawn from the bank because lower balance will earn lower interest.

If there are Q number of visits to bank and cost of per visit (or transaction cost) is b, then the cost of visit should be :

$$bQ$$

Moreover, if i is the interest rate, then the opportunity cost should be

$$\frac{iY}{2Q}$$

Therefore, total cost (TC) should be

$$TC = \frac{iY}{2Q} + bQ$$

Now, to minimise the total cost, we use differential calculus and find out the first-order condition of minimization i.e.,

$$\frac{d\varphi}{dQ} = 0$$

$$-\frac{iY}{2Q^2} + b = 0$$

$$b = \frac{iY}{2Q^2}$$

$$Q^2 = \frac{iY}{2b}$$

$$Q = \sqrt{\frac{iY}{2b}}$$

Further, the average money holding is

$$A = \frac{Y}{2Q} = \frac{1}{2} \times \frac{Y}{\sqrt{\frac{iY}{2b}}}$$

$$A = \frac{1}{2} \times Y\sqrt{\frac{2b}{iY}}$$

$$A = \sqrt{\frac{bY}{2i}}$$

The above formula gives us the average money holding as per the Baumol-Tobin theory. The following illustrates the formula: Mr. M wants to withdraw from his bank balance of ₹ 1,00,000. The interest rate is 5% per annum and transaction cost (or cost of per visit) ₹ 4, then the optimum number of visits and average money holding can be calculated as follows:

$$Q = \sqrt{\frac{0.05 \times 1{,}00{,}000}{2 \times 4}} = 25$$

$$A = \sqrt{\frac{4 \times 1{,}00{,}000}{2 \times 0.05}} = \sqrt{40{,}00{,}000} = 2{,}000$$

The above results can be verified through the following table and Figure 2. This is obvious that total cost curve TC is minimum at Q=25.

Number of visits (Q)	**Average monthly holding = Y/2Q**	**Visit cost = Q×₹4**	**Opportunity cost = Average monthly holding × 5%**	**Total Cost**
5	10000	20	500.00	520.00
7	7143	28	357.14	385.14
10	5000	40	250.00	290.00
13	3846	52	192.31	244.31
16	3125	64	156.25	220.25
19	2632	76	131.58	207.58

22	2273	88	113.64	201.64
25	**2000**	**100**	**100.00**	**200.00**
28	1786	112	89.29	201.29
31	1613	124	80.65	204.65
34	1471	136	73.53	209.53
37	1351	148	67.57	215.57
40	1250	160	62.50	222.50
43	1163	172	58.14	230.14
46	1087	184	54.35	238.35
49	1020	196	51.02	247.02

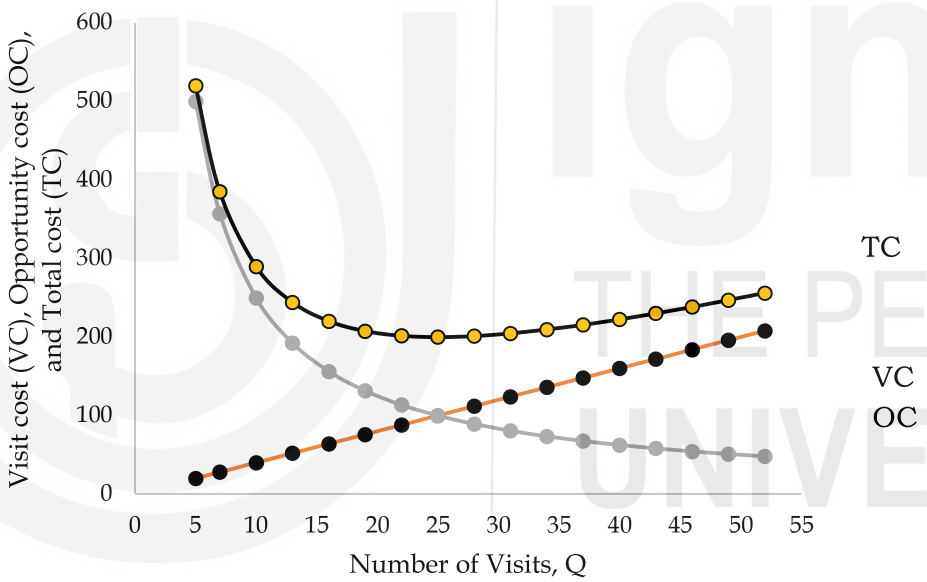

Fig. 5.2 : Optimum Number of Visits

Criticisms to the Baumol – Tobin Theory

a) The theory assumes that the money is spent at constant rate, but in real life this assumption is unrealistic because people do not know with certainty how much they need to spend.

b) It is not always easy to quantify the cost of transactions.

5.4.5 Tobin's Theory of Speculative Demand for Money

James Tobin published a paper[2], in which he says that individuals make the portfolio of money as well as bond, and the speculative demand for money is

[2] Tobin, J. (1958). Liquidity Preference as Behavior Towards Risk. The Review of Economic Studies, 25(2), 65. https://doi.org/10.2307/2296205

a matter of uncertainty not the expectations about the interest rates as opposed to Keynes who used to think that individuals hold either all cash or all bonds depending upon their expectations of interest rate. Let us understand the view of Tobin. Suppose that V represents the value of the bonds and r is the rate of return, then total return should be

$$R = rV$$

Moreover, if θ is rate of risk, then the total risk should be

$$\delta = \theta V$$

Combining these two equations we get

$$R = \frac{r}{\theta}\delta$$

The ratio $\frac{r}{\theta}$ should be treated as a constant β , then we have

$$R = \beta\delta$$

The last equation shows a linear relationship between R and δ as shown in Figure 5 through the line OZ. The line shows that higher the level of risk (shown on the horizontal axis in panel (a)) leads higher return (shown on the vertical axis in panel (a)).

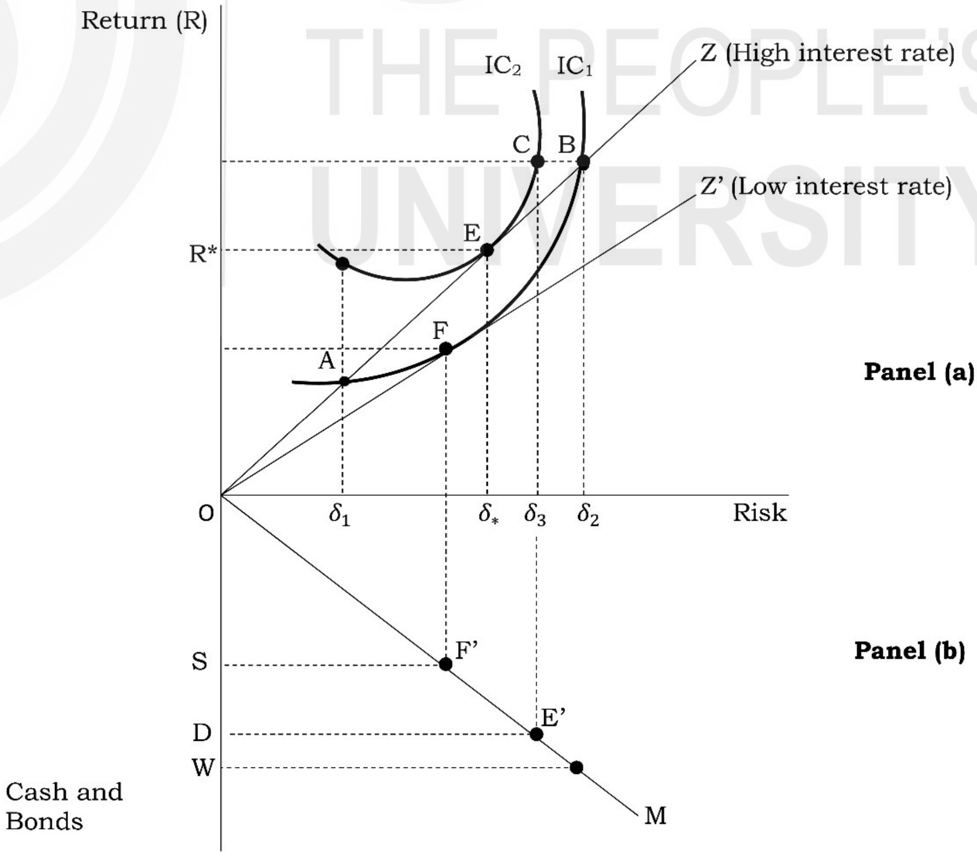

Fig. 5.3: Tobin's Portfolio Optimization and Bonds and Cash Weights Determination

In panel (a), IC_1 and IC_2 are two indifference curves showing different levels of utility for various combinations of risk and return. From the properties of the indifference curve, we know that higher the level of indifference curve is preferable, therefore, an investor would prefer IC_2 in comparison to IC_1. To prove this, let us consider that the investor choses point A on IC_1 where the risk is δ_1, but with the same amount of risk, a higher return on IC_2 is available, but that return is not achievable because it does not fall on OZ. Similarly, at point B, the risk is δ_2, but the same return at point C on IC_2 is available with lower risk δ_3 . However, point C cannot be achieved because it also does not fall on OZ. Thus, any point below IC_2 will give the investor lower utility because of the opportunity to achieve the higher indifference curve. Thus, the investor will maximise his utility when the OZ line touches the indifference curve at point E. To achieve this point E, the investor would need investment in bonds equal to OD out of total wealth W as shown on the vertical axis in panel (b). Thus, remaining amount WD is cash remaining with the investor. The line OM shows such various combinations of bond and cash available with the investor for a given point in panel (a). Now, if the interest rate falls, then the OZ line will rotate clockwise, say, OZ' touching IC_1 at point F. Thus, new equilibrium (point of optimization) point is F at which investment in bonds is OS while the cash is WS. This is obvious that OS < OD implying that fall in interest rate causes the investment in bonds to fall. Therefore, according to Tobin, people combine bonds and cash in a ratio to make a portfolio depending upon their point of optimization.

5.4.6 Friedman's Quantity Theory of Money

Milton Friedman, an American economist cum statistician and the 1976 Nobel prize winner, developed his theory of demand for money. Infact, Friedman regarded his theory as a restatement of the quantity theory of money, that is why, his theory sometimes called modern quantity theory of money. According to Friedman, money demand can be studied in the same manner as the demand for any asset. Thus, the Friedman's 1970 version of the demand for money function is as follows:

$$M_d = f(Y_P, W,\ i_m,\ i_b,\ i_e, \pi_e, u)P$$

where,

P is the price level. The logic of inclusion of the price level is that changes the purchasing power of the money. If P increases, then it reduces the real money balance i.e., the purchasing power and to compensate it people demand more money.

Y_P is permanent income which is used as a proxy of human wealth (i.e., labour earnings) because wealth cannot be directly measured. Permanent income is weighted average of the past and present incomes. Higher the level of permanent income means higher the money demand.

W is non-human wealth e.g., land or machines.

i_m is the return on money itself and if it increases, then the people demand more money.

i_b is the return on bond. Its decrease or increase result in capital gain or loss, therefore, it is negatively related to the money demand.

i_e is the return on equities. It is negatively related to money demand because higher the return on equities will attract the investors to earn dividends and, therefore, less money demand.

π_e is the expected inflation rate. When people expect inflation in future, then their purchasing power will decline, therefore, they would like spend or invest money before their expectations come true. Therefore, there is a negative relationship between the money demand and expected inflation rate. Expected inflation can be calculated as follows:

$$\pi_e = \frac{1}{P}\frac{dP^e}{dt}$$

u is a variable representing all those factors not considered in the function e.g., taste and preferences. Its relationship with the money demand may be in either of the directions.

Check Your Progress 2

1) Although the quantity theory of money is not a theory of demand, then how can it be used as a theory of demand for money?

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2) What are the three motives for which money is demanded ?

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3) What is Tobin's theory for demand for money?

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4) What is the usefulness of the Baumol – Tobin model?

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5) Explain the Friedman's quantity theory of money.

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5.5 LET US SUM UP

Money demand refers to the desire of individuals and institutions to hold cash and other liquid assets for various purposes. It plays a crucial role in determining the overall demand for money within an economy, which in turn influences monetary policy decisions and economic stability.

Factors Affecting Money Demand

Interest rate : Interest rate is the price of demanding money. Holding money carries an opportunity cost in the form of interest lost which could be earned if the money demanded were invested in some other assets like bond. Thus, there is a negative relationship between the demand for money and interest rate.

Income (or GDP) : Higher the level of income means higher the volume of transactions, therefore, there is a positive relationship between the income and demand form money.

Price Level: The price level is a significant determinant of money demand. As prices rise, the purchasing power of money decreases, leading individuals to demand more money to maintain their desired level of transactions. Conversely, when prices fall, the demand for money decreases as individuals require less cash for the same level of transactions.

Risk associated in alternative assets : If the alternative assets are risky, then the demand for money.

Technology of Payment and Receipt: Advancements in payment technologies, such as digital payment platforms like Paytm, can affect money demand. These technologies offer convenient and efficient alternatives to cash transactions, potentially reducing the need for holding physical currency. As electronic payment methods become more widespread and accessible, individuals may hold less cash, impacting money demand.

Theories of Money Demand

Quantity Theory of Money (Fisher Equation): This theory posits a direct relationship between the quantity of money in circulation and the price level in an economy. According to the Quantity Theory of Money, formulated by Irving Fisher, MV = PT, where M represents the money supply, V denotes the velocity of money (the rate at which money circulates), P stands for the price level, and T represents the volume of transactions. The theory suggests that changes in the money supply directly impact the price level, assuming a stable velocity of money and volume of transactions. The theory does not specifically talk about the demand for money, but it helps us to find out the demand in equilibrium situation.

Cambridge Version of Quantity Theory of Money: The Cambridge version of the Quantity Theory of Money emphasizes the role of the demand for money in determining the price level. The theory establishes that the transaction demand for money is a fraction of nominal income. It can be used to find out the money supply in equilibrium situation. The theory is also known as neoclassical theory of demand for money.

Liquidity Preference Theory of Money (Keynesian Perspective): Proposed by John Maynard Keynes, the Liquidity Preference Theory of Money focuses on the demand for money as a store of wealth rather than solely a medium of exchange. According to this theory, individuals hold money not only for transactional purposes but also as a precautionary measure against uncertainty and for speculative purposes.

Baumol – Tobin Theory for Transactive Purpose: This theory helps to find out the optimum level of money to minimise the total cost made of transaction cost and opportunity cost.

Tobin's Theory for Speculative Money demand: This theory establishes a negative relationship between the interest rate and demand for money (i.e., cash) and positive relationship between the interest rate and the demand for bonds. According to Tobin, people hold cash and bonds both as oppose to Keynes whose view is that people either hold all cash or all bonds.

Friedman's Quantity Theory of Money : The theory sees money as an asset explains its demand in the same manner as the demand for any other good is explained.

5.6 KEY WORDS

Bullish nature : A bullish nature refers to a positive or optimistic outlook on the future performance of an asset or market. It suggests an expectation of rising prices and a favourable market environment.

Bearish nature : A bearish nature describes a negative or pessimistic outlook on the future performance of an asset or market. It implies an expectation of declining prices and an unfavourable market environment.

Bond : A bond is a debt investment where an investor loans money to an entity (typically a corporation or government) which borrows the funds for a defined period at a variable or fixed interest rate.

Capital gain : Capital gain refers to the profit earned from the sale of an asset, such as stocks, real estate, or bonds, which exceeds its purchase price.

Transactive motive for money demand : This motive refers to the need for money to facilitate day-to-day transactions such as buying goods and services. Individuals and businesses hold money to carry out their regular spending activities.

Precautionary motive for money demand : The precautionary motive for money demand refers to the desire to hold money as a buffer or reserve for unexpected future expenses or emergencies. It serves as a precaution against unforeseen circumstances.

Speculative motive for money demand : This motive involves holding money not for transactional purposes or precautionary reasons, but rather for potential investment opportunities. Individuals may hold money to take advantage of favorable market conditions or to quickly seize investment opportunities.

Velocity of money : Velocity of money is the rate at which money is exchanged in an economy. It is usually measured as the number of times money is spent in a given period. High velocity indicates a rapid circulation of money, while low velocity suggests slower economic activity.

5.7 SOME USEFUL REFERENCES

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5.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1. Money demand means the quantity of money (or cash) which households and firms want to hold with themselves, but not in other forms like bonds. Keeping other factors constat, just as the demand for a good is negatively related with its price, similarly, the demand for money is negatively related with interest rate.

2. Refer to Section 5.2

Check Your Progress 2

1. Refer the relationship between the quantity theory of money and demand for money in section 5.4.

2. Transactive motive, precautionary motive, speculative motive

3. Individuals make the portfolio of money as well as bond, and the speculative demand for money is a matter of uncertainty not the expectations about the interest rates as opposed to Keynes who used to think that individuals hold either all cash or all bonds depending upon their expectations of interest rate.

4. Refer to Section 5.4.4

5. According to Friedman, money demand can be studied in the same manner as the demand for any asset. The Friedman's 1970 version of the demand for money function: $M_d = f(Y_P, W,\ i_m,\ i_b,\ i_e, \pi_e, u)P$

UNIT 6 MONEY SUPPLY

Structure

6.0 OBJECTIVES

After going through the unit, you should be able to:

- Define high-powered money;
- Explain the link between high-powered money and the money supply;
- Explain the money-multiplier process;
- List the determinants of the money multiplier;
- Describe the factors which affect high-powered money; and
- Discuss the theory of endogenous money supply.

6.1 INTRODUCTION

In units 4 and 5, you learned about the nature and functions of money, and about the demand for money. In this unit, we will discuss supply of money. Till now, we have implicitly assumed that the supply of money is policy-determined. In an actual economy, however, the supply of money is determined jointly by the monetary authority and banks, and also in a way by the public. The central role in this is played by the monetary authority. We first need to be clear about the measure of money. Let us consider the narrow measure of money that we had mentioned in unit 4. That was M1 which was a sum of the currency and demand deposits. In this unit, we shall make use of another measure of money, but money that is created by the monetary authorities (Central Bank and the Government). This is called high-powered money.

In this unit, the discussion centres on how high-powered money, through a 'money-multiplier', determines the money supply. The unit discusses the nature of the high-powered money and its components. A discussion is provided about the nature of multiplier, its components and the relation among the components of the multiplier. You will come to understand why the multiplier is called 'multiplier'. The unit discusses the factors which determine the money multiplier and high powered money. Finally, the unit discusses some approaches and views that suggest that money supply is not exogenous but is itself endogenously determined.

6.2 HIGH POWERED MONEY AND MONEY SUPPLY

In the realm of macroeconomics and monetary policy, the concepts of high-powered money and money supply play pivotal roles. High-powered money, also known as the monetary base or monetary base supply, represents the total amount of a country's currency in circulation along with its reserves held by the central bank. Understanding high-powered money is crucial for comprehending the mechanisms through which central banks influence the broader money supply in an economy. This section aims to elucidate the concept of high-powered money, its components, and its implications on the broader money supply.

Let us study relation between high powered money and money supply with the help of the following equations:

$$M = C + D \tag{1}$$

where M= Money supply; C = Currency; D= Demand Deposit

$$H = C + R \tag{2}$$

$$\frac{H}{MB} = C + [RR + ER] \tag{3}$$

$$c^d = \frac{C}{D} \text{ or } C = c^d * D \tag{4}$$

$$rr = \frac{RR}{D} \text{ or } RR = rr * D \tag{5}$$

$$er = \frac{ER}{D} \text{ or } ER = er * D \tag{6}$$

where H = High Powered Money; C= Currency; R= Reserves; RR = Required Reserve; ER=Excess Reserve; MB= Monetary Base; c^d = Currency to Deposit ratio ; rr = required reserve to deposit ratio ; er = excess reserves to deposit ratio ; m= money multiplier

$$c^d = \frac{C}{D} \text{ or } C = c^d * D$$

$$rr = \frac{RR}{D} \text{ or } RR = rr * D$$

$$er = \frac{ER}{D} \text{ or } ER = er * D$$

$$M = m * H \text{ or } m = \frac{M}{H} \qquad (7)$$

$$\boldsymbol{m = \frac{C + D}{C + [RR + ER]}}$$

Dividing the above equation by D we get:

$$m = \frac{\frac{C}{D} + \frac{D}{D}}{\frac{C}{D} + \frac{RR}{D} + \frac{ER}{D}} \qquad (8)$$

$$m = \frac{c^d + 1}{c^d + rr + er} \qquad (9)$$

$$since\ m = \frac{M}{H}$$

$$\frac{M}{H} = \frac{c^d + 1}{c^d + rr + er} \qquad (10)$$

M/H = c^d +1/ c^d +rr + er **(11)**

If there is no c^d;

$$m = \frac{1}{rr + er}$$

If there is no c^d and er;

$$m = \frac{1}{rr}$$

Components of High-Powered Money:

High-powered money comprises two main components:

1) **Currency in Circulation:** This component encompasses all the physical currency (notes and coins) circulating within an economy. It includes the money held by individuals, businesses, and institutions such as banks.

2) **Reserves Held by the Central Bank**: Reserves refer to the deposits that commercial banks hold at the central bank. These reserves are mandatory and serve as a buffer against financial crises and liquidity shortages. There are two types of reserves:

— *Required Reserves:* Mandated by central banks, these reserves are a fraction of a bank's deposits that it must hold in reserve either as cash in its vaults or as deposits with the central bank.

— *Excess Reserves:* These are reserves held by banks over and above the required amount. Banks may hold excess reserves for various reasons, including precautionary measures, meeting withdrawal demands, or regulatory compliance.

Implications of High-Powered Money on Money Supply

Understanding the relationship between high-powered money and the broader money supply is essential for comprehending monetary policy and its effects on the economy. The money supply refers to the total amount of money circulating in the economy, which includes both physical currency and various forms of bank deposits.

The process through which high-powered money influences the broader money supply can be elucidated through the money multiplier mechanism, which operates as follows:

1) **Money Multiplier Mechanism**: The central bank controls the monetary base through its open market operations, reserve requirements, and discount rates. By altering the monetary base, the central bank influences the overall money supply through a process called the money multiplier effect. The money multiplier represents the ratio of the change in the money supply to the change in the monetary base.

2) **Reserve Requirements**: When the central bank alters reserve requirements, it directly affects the amount of excess reserves that banks hold. Lowering reserve requirements increases excess reserves, providing banks with more capacity to extend loans, thereby expanding the money supply. Conversely, raising reserve requirements reduces excess reserves, limiting banks' lending capacity and contracting the money supply.

3) **Open Market Operations**: Through open market operations, the central bank buys or sells government securities in the open market. When the central bank purchases government securities, it injects funds into the banking system, increasing reserves and enabling banks to lend more, thus expanding the money supply. Conversely, selling government securities reduces reserves, constraining banks' ability to lend and contracting the money supply.

4) **Discount Rate**: The discount rate is the interest rate at which commercial banks can borrow from the central bank. When the central bank lowers the discount rate, borrowing becomes cheaper for commercial banks, encouraging them to borrow more, increase lending, and expand the money supply. Conversely, raising the discount rate

makes borrowing more expensive, leading to a decrease in lending and a contraction of the money supply.

High-powered money, comprising currency in circulation and reserves held by the central bank, serves as the foundation for the broader money supply in an economy. Understanding the relationship between high-powered money and the money supply is crucial for policymakers, economists, and students alike. By manipulating the components of high-powered money through various monetary policy tools, central banks can influence the money supply, thereby affecting economic activity, inflation, and interest rates. Thus, a thorough comprehension of high-powered money is essential for analyzing and formulating effective monetary policies.

Check Your Progress 1

1) What are the two main components of high-powered money?

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2) How does the central bank influence the money supply through open market operations?

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6.3 THE MONEY MULTIPLIER PROCESS

The money multiplier process is a fundamental concept in monetary economics, elucidating the mechanisms through which changes in the monetary base, controlled by central banks, influence the broader money supply in an economy. This section aims to provide a comprehensive understanding of the money multiplier process, its determinants, and its implications for monetary policy and economic stability.

Complete Deposit Multiplier

From equation (3), we know $\frac{H}{MB} = C + [RR + ER]$

$$or\ MB = c^d * D + rr * D + er * D$$

$$or\ MB = D[c^d + rr + er]$$

$$or\ D = \frac{MB}{c^d + rr + er}$$

Here $\frac{1}{c^d + rr + er}$ is complete deposit multiplier

$$If\ c^d = 0\ and\ er = 0\ then\ \frac{1}{rr}\ is\ simple\ deposit\ multiplier$$

where MB= Monetary Base; c^d = Currency to Deposit ratio; rr = required reserve to deposit ratio; er = excess reserves to deposit ratio; D = Demand deposit

Complete Currency multiplier

$$c^d = \frac{C}{D}$$

$$C = c^d * D$$

We know from above that:

$$D = \frac{MB}{c^d + rr + er}$$

$$C = c^d * \frac{MB}{c^d + rr + er}$$

$$= \left[\frac{c^d}{c^d + rr + er}\right] * MB$$

Here $\frac{c^d}{c^d + rr + er}$ is a complete currency multiplier

Where MB= Monetary Base; c^d = Currency to Deposit ratio; rr = required reserve to deposit ratio; er = excess reserves to deposit ratio; D = Demand deposit

The Concept of the Money Multiplier

The money multiplier represents the ratio of the change in the money supply to the change in the monetary base. It illustrates how an initial injection of high-powered money (base money) into the banking system leads to a multiple expansion of the broader money supply through the process of bank lending and deposit creation.

Mechanisms of the Money Multiplier Process

The money multiplier process operates through a series of steps, as outlined below:

1) **Initial Injection of Base Money:** The process begins with the central bank injecting base money into the banking system through open market operations, discount window lending, or other monetary policy tools.

2) **Reserve Requirement:** Commercial banks are required to hold a certain percentage of their deposits as reserves, either as vault cash or as deposits with the central bank. This reserve requirement acts as a constraint on the amount of money banks can create through lending.

3) **Deposit Creation through Lending:** When banks receive additional reserves, either through central bank injections or deposit inflows, they can expand their lending activities. Loans create new deposits, effectively increasing the money supply beyond the initial injection of base money.

4) **Multiplier Effect:** The expansion of bank loans and deposits sets off a multiplier effect as the newly created deposits become a part of the broader money supply. These deposits are re-deposited in banks or spent in the economy, leading to further rounds of lending and deposit creation.

5) **Limitations and Leakages:** The money multiplier process is subject to limitations and leakages that can dampen its impact on the money supply. These include excess reserve holdings by banks, currency drain (withdrawal of currency from banks), and non-bank financial intermediaries that do not fall under traditional reserve requirements.

Determinants of the Money Multiplier

Several factors influence the magnitude of the money multiplier and its effectiveness in expanding the money supply:

1) **Reserve Ratio:** The reserve requirement set by the central bank directly affects the money multiplier. A lower reserve ratio allows banks to lend out more of their deposits, leading to a higher multiplier effect.

2) **Currency Drain:** The propensity of the public to hold currency rather than deposits affects the effectiveness of the money multiplier. A higher currency drain reduces the deposit base on which banks can create new loans, thus lowering the multiplier.

3) **Excess Reserves:** The level of excess reserves held by banks influences their lending capacity. Higher excess reserves can limit the effectiveness of the multiplier process as banks may choose to hoard reserves rather than extend loans.

Implications for Monetary Policy and Economic Stability

Understanding the money multiplier process is crucial for central banks in formulating and implementing monetary policy. By adjusting the monetary base through open market operations, discount rate changes, or reserve requirement adjustments, central banks can influence the money supply and thereby regulate economic activity, inflation, and interest rates.

In conclusion, the money multiplier process serves as a cornerstone of monetary economics, providing insights into the mechanisms through which changes in the monetary base propagate through the financial system to impact the broader money supply. A nuanced understanding of the money multiplier process is essential for policymakers, economists, and students alike, as it underpins the formulation and implementation of effective monetary policy measures aimed at achieving macroeconomic stability and sustainable economic growth.

Check Your Progress 2

1) What is the role of the reserve requirement in the money multiplier process?

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2) How does the money multiplier process contribute to the expansion of the broader money supply?

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3) Why is understanding the money multiplier process crucial for central banks in formulating monetary policy?

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6.4 FACTORS AFFECTING THE MONEY MULTIPLIER AND HIGH POWERED MONEY

Understanding the dynamics of money supply is crucial for central banks and policymakers in managing monetary policy effectively. The money multiplier and high powered money are key concepts in this regard. The money multiplier represents the ratio of the money supply to the monetary base, while high powered money refers to the monetary base itself. Various factors influence both the money multiplier and high powered money, which in turn impact the overall money supply in an economy.

Factors Affecting the Money Multiplier:

1) **Reserve Requirements:** The reserve requirement ratio set by the central bank significantly affects the money multiplier. A lower reserve requirement leads to a higher money multiplier, as banks can lend out more of their deposits, increasing the money supply.

2) **Currency Drain:** When individuals hold more currency rather than depositing it in banks, the money multiplier decreases. This is because

currency held by the public does not contribute to the creation of new deposits through the banking system.

3) **Banking Behaviour:** The behaviour of banks also affects the money multiplier. If banks become more cautious and hold excess reserves rather than lending them out, the money multiplier decreases. Conversely, if banks are more aggressive in lending, the money multiplier increases.

4) **Central Bank Policy:** Monetary policy actions by the central bank, such as open market operations and changes in the discount rate, can influence the money multiplier. For example, when the central bank purchases government securities in open market operations, it injects reserves into the banking system, potentially increasing the money multiplier.

5) **Deposit Leakage:** Leakage of deposits from the banking system, such as through excess reserves, excess currency holdings, or withdrawals for purchases of securities or assets outside the banking system, reduces the effectiveness of the money multiplier.

Factors Affecting High Powered Money:

1) **Central Bank Operations:** High powered money is primarily influenced by the actions of the central bank. Central banks control the monetary base through various operations, including open market operations, changes in reserve requirements, and lending facilities.

2) **Government Transactions:** Government transactions, such as fiscal policy decisions that involve changes in government spending or taxation, can impact high powered money. For example, government expenditures financed by the central bank may increase the monetary base.

3) **Foreign Exchange Operations:** Foreign exchange interventions by central banks can affect high powered money, especially in economies with fixed or managed exchange rate regimes. Purchases or sales of foreign currency reserves impact the monetary base.

4) **Financial Crises:** During financial crises, central banks may engage in unconventional measures such as quantitative easing, which involves increasing the monetary base through large-scale purchases of financial assets.

5) **Banking Sector Dynamics:** Changes in the banking sector, such as the consolidation of banks or shifts in the composition of bank reserves, can influence high powered money.

The money multiplier and high powered money are interconnected concepts that play a crucial role in determining the overall money supply in an economy. Understanding the factors that influence these variables is essential for central banks and policymakers in formulating and implementing effective monetary policy. By monitoring and managing these factors, authorities can work to maintain price stability, promote economic growth, and achieve their policy objectives.

Check Your Progress 3

1) How does a decrease in reserve requirements affect the money multiplier?

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2) What impact does excess currency holdings by the public have on the money multiplier?

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3) Why is it important for central banks to consider government transactions when assessing high powered money?

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6.5 THE THEORY OF ENDOGENOUS MONEY SUPPLY

The theory of endogenous money supply challenges the traditional view of money creation, which suggests that the money supply is primarily determined by the actions of central banks. Instead, it posits that the money supply is largely endogenously determined by the demand for credit from borrowers and the ability of banks to create money through the process of lending. This theory has significant implications for understanding the dynamics of monetary policy, financial stability, and economic activity.

Key Concepts:

1) Credit Creation by Banks

In the endogenous money supply framework, banks play a pivotal role in money creation through the process of credit creation. When banks extend loans to borrowers, they simultaneously create new deposits in the accounts of the borrowers. These deposits effectively function as money in the economy, increasing the overall money supply.

2) Loanable Funds vs. Money Supply

Unlike the loanable funds theory, which suggests that banks act as intermediaries that lend out pre-existing deposits, the endogenous money supply theory emphasizes that banks create new money when they make loans. Thus, the money supply is not limited by the availability of pre-existing funds but is determined by the demand for credit and the lending capacity of banks.

3) Reserves and Central Bank Influence

While central banks retain some influence over the money supply through their control of the monetary base (reserves and currency), the endogenous money supply theory suggests that this influence is indirect. Central bank actions, such as setting interest rates or conducting open market operations, affect the cost of borrowing and the willingness of banks to extend credit, thereby influencing the money supply.

4) Creditworthiness and Economic Conditions

The ability of banks to create credit and expand the money supply is contingent upon the creditworthiness of borrowers and prevailing economic conditions. In times of economic expansion and confidence, banks may be more willing to lend, leading to an increase in the money supply. Conversely, during economic downturns or financial crises, banks may become more cautious, leading to a contraction in credit and the money supply.

5) Liquidity Preference and Monetary Policy Transmission

The endogenous money supply theory highlights the role of liquidity preference and interest rate expectations in monetary policy transmission. Changes in central bank policy rates influence the cost of borrowing and the demand for credit, thereby impacting the money supply and overall economic activity.

Graphical Representation of the Exogenous and Endogenous Money Supply Curves

We further discuss the difference between the exogenous and endogenous money supply curve with the help of Figure 6.1 whereby part A shows the exogenous supply curve and the factors affecting the shift in the exogenous money supply curve. Part B discusses the endogenous supply curve and the factors affecting the shift in the endogenous supply curve.

We discussed the following terms in Section 6.3:

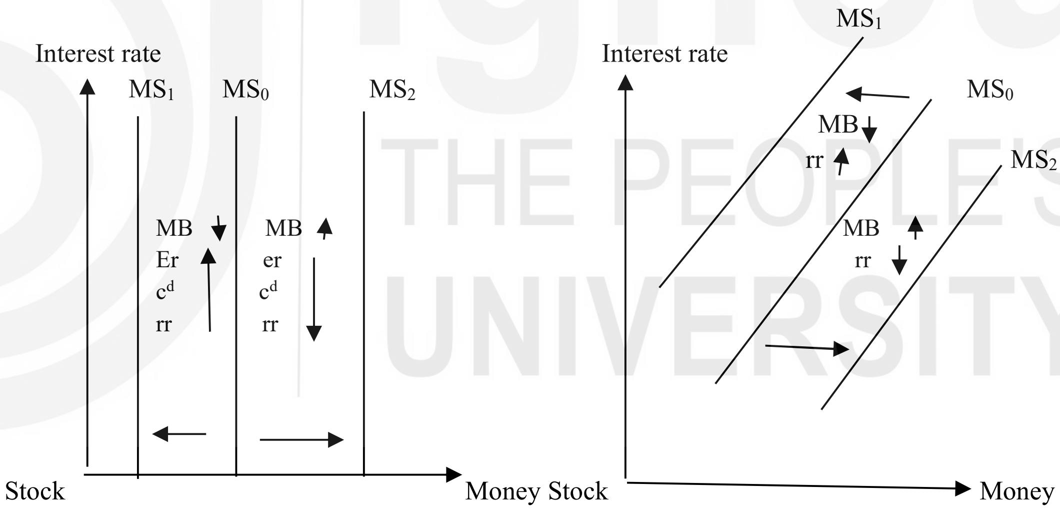
MB= Monetary Base; c^d = Currency to Deposit ratio; rr = required reserve to deposit ratio; er = excess reserves to deposit ratio; D = Demand deposit. These terms will be further used in the discussion.

Money supply curve shows the amount of money that suppliers are willing and able to supply at various interest rates in the economy. We know that there are four determinants of money supply, that is, MB, rr, er, and cd. From equation (10) from section 6.2, we know

Money supply=[c^d +1/ c^d +rr + er] * MB

But interest rates have no relation with MB and rr because it is determined by the Central Bank. Interest rates have impact on er and c^d. Economists differ in their views whereby some say that interest rate has no relation with er and c^d (Exogenous Money Supply Curve). Others say that interest rate has relation with er and c^d (Endogenous Money Supply).

In Figure 6.1 (A), we see that any change in the interest rates have no impact on the shape of the money supply curve, it is a vertical straight line (constant), say MS_0. As per The endogenous money supply curve in Figure 6.1 (B) we see that an increase in interest rates by the Central Bank Leads to more loans by the commercial banks in the market and hence increased money supply. So there is a positive relation between interest rates and money supply, upward sloping MS_0 curve in Figure 6.1 (B). At higher interest rates the customers prefer to deposit more money with the commercial banks to earn interest (c^d decreases), hence money available for loans increases with the commercial banks leading to increased money supply. Hence er and c^d impact money supply curve as per endogenous money supply surve.



A. Exogenous Money Supply Curve **B. Endogenous Money Supply Curve**

Fig. 6.1: Money Supply Curves and factors impacting shift in the curves

Further we discuss the shifts in both exogenous and endogenous money supply curve. In Figure 6.1 (A), we see that a decrease in MB and increase in er, c^d and rr leads to decrease in the money supply curve (leftwards), vice versa. Here it needs to be noticed that er and c^d are not impacted by interest rates, however, they do impact shifts in money supply.

In Figure 6.1 (B), we see that an increase in MB and rise in rr leads to increased money supply (rightward shift), MS_2 , vice versa. Here, er and c^d are impacted by interest rates and hence impact the money supply curve (upward sloping).

Implications:

1) Monetary Policy Effectiveness

Endogenous money supply theory suggests that monetary policy effectiveness may be influenced by the responsiveness of banks and borrowers to changes in central bank policy. Understanding the dynamics of credit creation and money supply determination is crucial for policymakers in assessing the impact of monetary policy on the economy.

2) Financial Stability Considerations

The theory of endogenous money supply emphasizes the interconnectedness of the banking system and the real economy. Financial stability considerations, such as the adequacy of bank capital and the management of systemic risks, are essential for ensuring the smooth functioning of the endogenous money supply mechanism.

3) Role of Central Banks

While central banks retain a significant role in influencing monetary conditions and financial stability, the endogenous money supply theory underscores the importance of understanding the behaviour of banks and borrowers in shaping the overall money supply dynamics.

The theory of endogenous money supply provides a different perspective on the process of money creation and the role of banks in the economy. By highlighting the endogenous nature of the money supply and the importance of credit creation by banks, this theory deepens our understanding of monetary policy transmission, financial stability, and the functioning of the monetary system. Incorporating these insights into economic analysis and policymaking can lead to more effective strategies for managing monetary conditions and promoting sustainable economic growth.

Check Your Progress 4

1) How does the theory of endogenous money supply differ from the traditional view of money creation?

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2) How do changes in central bank policy rates influence the money supply according to the theory of endogenous money supply?

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6.6 LET US SUM UP

The foundation of any economy lies in its monetary framework. Understanding the nuances of money supply is imperative for policymakers and economists alike. At the core of this understanding is high-powered money, which forms the basis for the broader monetary aggregates. Central banks wield significant control over the money supply through various tools at their disposal. By manipulating the monetary base through operations such as open market transactions, reserve requirement adjustments, and changes in the discount rate, central banks can influence economic activity, inflation, and interest rates.

The money multiplier process illustrates how changes in the monetary base trickle down through the financial system, amplifying the overall money supply. This process begins with an injection of base money into the banking system, leading to the creation of deposits as commercial banks extend loans. The subsequent expansion of bank loans and deposits triggers a multiplier effect, resulting in a proportional increase in the money supply. However, the effectiveness of this process is subject to various factors, including reserve requirements, currency drain, and the level of excess reserves held by banks.

In contrast to traditional views of money creation, the theory of endogenous money supply sheds light on the crucial role of banks in determining the money supply through credit creation. According to this theory, money is generated when banks extend loans to borrowers, thereby contributing to the expansion of the money stock. Understanding the intricacies of credit creation and the behaviour of banks and borrowers is essential in shaping our comprehension of money supply dynamics.

Overall, this exploration provides a comprehensive overview of the mechanisms governing money supply, highlighting the roles of high-powered money, the money multiplier process, and the theory of endogenous money supply. By delving into these concepts and their implications for monetary policy and economic stability, stakeholders can deepen their understanding of macroeconomic phenomena and make informed decisions to steer economies towards sustainable growth and stability.

6.7 KEY WORDS

High-Powered Money : The total amount of currency in circulation and reserves held by the central bank, serving as the foundation for the broader money supply in an economy.

Money Multiplier : A ratio representing the change in the money supply relative to the change in the monetary base, illustrating how an initial injection of high-powered money leads to a multiple expansion of

		the broader money supply through bank lending and deposit creation.
Exogenous	:	External or outside influence; in the context of money supply, exogenous refers to factors that are determined externally, such as by central banks.
Endogenous Money Supply	:	A theory challenging the traditional view of money creation, suggesting that the money supply is largely determined by the demand for credit from borrowers and the ability of banks to create money through lending, rather than by central bank actions alone.
Monetary Base	:	Synonymous with high-powered money, it represents the total amount of currency in circulation and reserves held by the central bank.
Open Market Operations	:	Actions undertaken by central banks to buy or sell government securities in the open market, influencing the amount of reserves in the banking system and thereby impacting the money supply.
Reserve Requirements	:	Mandated reserves that commercial banks are required to hold by central banks, affecting the amount of excess reserves available for lending and thus influencing the money supply.
Liquidity Preference	:	The desire of individuals and businesses to hold assets in liquid form, such as cash or demand deposits, rather than in illiquid forms such as physical assets or long-term investments.

6.8 SOME USEFUL BOOKS

Errol D'Souza (2012) Macroeconomics 2nd edition Pearson Education, New Delhi

McCallum, Benett Monetary Economics

6.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1. Currency and reserves.

2. Through open market operations, the central bank buys or sells government securities in the open market.

Check Your Progress 2

1. Reserve requirement acts as a constraint on the amount of money banks can create through lending.

2. Refer to Section 6.3

3. Refer to Section 6.3

Check Your Progress 3

1. A lower reserve requirement leads to a higher money multiplier, as banks can lend out more of their deposits, increasing the money supply.

2. When individuals hold more currency rather than depositing it in banks, the money multiplier decreases. This is because currency held by the public does not contribute to the creation of new deposits through the banking system.

3. Government transactions, such as fiscal policy decisions that involve changes in government spending or taxation, can impact high powered money.

Check Your Progress 4

1. Refer to Section 6.5

   While central banks retain some influence over the money supply through their control of the monetary base (reserves and currency), the endogenous money supply theory suggests that this influence is indirect. Central bank actions, such as setting interest rates or conducting open market operations, affect the cost of borrowing and the willingness of banks to extend credit, thereby influencing the money supply.